Shading-Aware AI System for Solar Investment, Health Monitoring, and Grid-Resilience in Sri Lanka

Proactive, Hardware-Free Grid-Survival Optimization for Solar Home Energy Systems

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Project Proposal Report

P K L Wijewardena – IT23334038

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology

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ABSTRACT

Sri Lanka's Ceylon Electricity Board periodically enforces scheduled load-shedding, cutting grid power to households at predictable intervals. While residential rooftop solar-battery systems are increasingly common, most rely on static threshold-based control with no awareness of upcoming outages. As a result, a household's battery may be critically low exactly when grid power is cut, leaving essential loads unserved. Existing research on home energy management largely optimizes for cost savings under normal grid conditions, or addresses unplanned outages caused by extreme weather. None specifically target scheduled, known-in-advance outages, and most assume access to smart-meter or inverter-level telemetry unavailable to most Sri Lankan households. This creates a clear research gap. There is no hardware-free, anticipatory battery control system designed for scheduled load-shedding, with the flexibility for households to balance cost savings against blackout resilience.

This research proposes a Model Predictive Control based decision system that determines hourly whether a household should self-consume export or reserve solar generated power, using known load shedding schedules and short-term generation and demand forecasts. A single Deep Reinforcement Learning agent is implemented as a secondary comparison method, to test whether a learning-based approach offers measurable advantage over calculated optimization for this problem. The study follows a simulation based experimental methodology, using real photovoltaic generation, load, and battery data from residential installations, combined with public schedule data from the Ceylon Electricity Board and weather data from the NASA POWER API. An iterative development approach refines the MPC formulation. Performance is evaluated against a rule-based baseline using total hours of unserved critical load and financial savings as primary metrics. This component forms the outage resilience layer of a broader four-part residential solar intelligence system, operating independently while remaining compatible with the system's shared architecture.

Keywords: Model Predictive Control, home energy management, load-shedding resilience, solar-battery optimization, residential energy resilience

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LIST OF ABBRIEVIATIONS

Abbreviation	Full Term
AI	Artificial Intelligence
API	Application Programming Interface
B2B	Business to Business
B2C	Business to Consumer

BESS	Battery Energy Storage System
CEB	Ceylon Electricity Board
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
ENMPC	Ensemble Nonlinear Model Predictive Control
EV	Electric Vehicle
FR	Functional Requirement
HEMS	Home Energy Management System
HPMS	Home Power Management System
IEEE	Institute of Electrical and Electronics Engineers
IP	Intellectual Property
IPR	Intellectual Property Rights
LCOE	Levelized Cost of Electricity
MDP	Markov Decision Process
MILP	Mixed Integer Linear Program
MPC	Model Predictive Control
MPPT	Maximum Power Point Tracking
NASA	National Aeronautics and Space Administration
NFR	Non-Functional Requirement
PPO	Proximal Policy Optimization
PV	Photovoltaic
SDG	Sustainable Development Goal
SMPC	Stochastic Model Predictive Control
SoC	State of Charge
WBS	Work Breakdown Structure

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Appendix 1: Mandatory Proposal Evidence and Decision Log

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1. INTRODUCTION

1.1. Background and Context

Sri Lanka has gone through repeated periods of scheduled power cuts, most severely during the 2022 economic crisis, when the Ceylon Electricity Board (CEB) imposed rolling blackouts of up to 13 hours a day because fuel shortages were hitting thermal generation [6]. Things have stabilized since then, with renewable sources now making up over 70% of national generation as of mid-2025 [7], but scheduled load-shedding is still a real risk whenever hydropower availability drops or there's a generation shortfall, and it's still carried out on a rotational, zone-based basis across the country [6].

At the same time, residential rooftop solar has grown fast, pushed along by government programs like Soorya Bala Sangramaya (Battle for Solar Energy) and schemes such as Net Metering, Net Accounting, and Net Plus [8]. Rooftop solar capacity went from about 750 MW in early 2024 to around 1,700 MW by May 2025 [8]. But most residential solar-battery systems still run on simple, static threshold-based logic that has no idea what the CEB's load-shedding schedule actually is. So a household's battery can end up sitting at a low charge right when a scheduled outage hits, leaving lighting, refrigeration, or medical equipment without power.

This matters for the growing number of Sri Lankan solar households, especially those in areas that see frequent rotational load-shedding. Battery storage is a significant financial investment, and right now its resilience benefit is left up to basic, non-adaptive control rules instead of being planned around outage timing that's already known in advance.

Existing home energy management research tackles related but different problems. Several studies use Model Predictive Control (MPC) to optimize battery dispatch for cost minimization under normal grid conditions [1], [3], [5], while others focus on resilience during unplanned outages caused by extreme weather [Gaikwad]. But these approaches either don't deal with outages at all, or deal with outages that are unpredictable rather than scheduled ahead of time. Many of them also assume access to smart-meter, inverter-level, or grid-tied market data [1], [2], [3], [5] that most Sri Lankan residential solar adopters simply don't have.

The proposed solution is an MPC-based decision system that uses the CEB's published load-shedding schedule, along with solar generation and household load forecasts, to work out hour by hour whether a household should self-consume, export, or reserve its solar power, all without needing extra hardware. A secondary Deep Reinforcement Learning (DRL) comparison is also included, to test empirically whether a learning-based approach actually beats calculated optimization for this specific, schedule-aware problem.

This work ties into UN Sustainable Development Goal 7 (Affordable and Clean Energy), by making existing residential solar investments more reliable and useful in practice, and SDG 9 (Industry, Innovation and Infrastructure), by helping build more resilient energy infrastructure at the household level. It sits within the Data Science research cluster, applying optimization and machine learning to a real-world energy systems problem.

1.2. Literature Review

Several recent studies have applied Model Predictive Control (MPC) to residential solar-battery energy management, though with different objectives, assumptions, and contexts.

[1] Saleem et al. (2024) - Stochastic MPC for Residential Complexes & EV Charging
Propose a Stochastic Model Predictive Control (SMPC) framework for multi-household residential complexes that combine rooftop PV, shared battery storage (BESS), and EV charging. They use SMPC over a 24-hour horizon with probabilistic scenario generation to capture correlated uncertainty in solar output, demand, and EV charging behavior, tested in MATLAB Simscape Electrical and validated in real time on an OPAL-RT simulator. The setting is grid-connected residential complexes with shared infrastructure. Their system improved battery longevity and profitability while keeping voltage stable at the point of common coupling. It handles multiple correlated uncertainty sources well and accounts for battery degradation and grid voltage support, but it assumes centralized control over shared BESS and EV infrastructure, and it optimizes for cost and grid stability rather than outage survival. No scheduled or unplanned outage is considered.

[2] Watil (2024) - Stand-Alone Off-Grid Home Power Management
Builds a real-time Home Power Management System (HPMS) for stand-alone, off-grid residential PV-battery systems, using five discrete operating modes (MPPT charging, partial charging, fast charging, float charging, partial discharging), each run by its own MPC. It's tested with real irradiance and temperature data through MATLAB/Simulink simulation and a hardware-in-the-loop setup. The context is homes with no grid connection at all. The mode-switching approach prevented battery over-voltage and over-current degradation while keeping power supply continuous under changing irradiance, and it's strong on low-level battery protection and real-time response. But it's built for permanently off-grid systems, not grid-connected households facing intermittent scheduled outages, and it doesn't touch export-to-grid decisions or outage anticipation.

[3] Li et al. (2023) - Ensemble Nonlinear MPC & Electrochemical Battery Wear
Present an Ensemble Nonlinear MPC (EnNMPC) for single-household prosumers in dynamic electricity markets, paired with a third-order physics-based electrochemical and thermal battery degradation model, and using historical error learning to manage forecast uncertainty. They run a year-long simulation on real-world prosumer load and price data, in a setting where households face volatile day-ahead market prices. The result was a 30%+ reduction in average daily operating cost compared to conventional mean nonlinear MPC, with detailed physics-based degradation modeling and solid handling of forecast error as its strengths. The catch is that it

assumes access to real-time market price signals and optimizes purely for cost, with no consideration of outage survival or critical load prioritization.

[4] Gaikwad et al. (2024) - Resiliency & Load Prioritization During Extreme Weather Outages

The most closely related work, develop an MPC-based Home Energy Management System (HEMS) for energy resilience during long, unplanned outages caused by extreme weather. They use a Mixed-Integer Linear Program (MILP) that balances thermal comfort (air conditioning) against a multi-level, circuit-grouped critical load priority stack, tested on real house load demand and weather data collected after an actual extreme weather event. The context is off-grid resilience during unplanned outages, evaluated against a rule-based controller and a baseline. Their MPC controller beat both in balancing thermal comfort with critical and other load service, especially when energy was constrained. It has a realistic load prioritization scheme and a strong resilience-focused evaluation, but the outages it handles are unplanned and weather-triggered rather than scheduled ahead of time. The system also assumes circuit-level smart panel control, referencing commercial hardware like SPAN.io, and offers no user-configurable resilience preference. Worth noting: the authors themselves flag a reinforcement-learning-based controller as unaddressed future work, since their MILP formulation is computationally heavy to solve.

[5] Kobou Ngani and Hadji-Minaglou (2023) - Profitability & Battery LCOE in Spot Electricity Markets

Look at how profitable residential battery storage is under volatile spot electricity prices, using a MILP-based MPC that folds the battery's Levelized Cost of Electricity (LCOE) directly into the optimization's cost function. They test it on day-ahead 15-minute spot market price data, for grid-connected homes optimizing financial return through self-consumption and feed-in tariffs. Including LCOE in the cost function stopped non-profitable battery cycling while preserving battery health, which directly ties battery economics to control decisions, a genuinely useful feature for real-world deployment. But it doesn't consider outage events, critical load, or resilience at all, and it assumes access to live market pricing data that doesn't exist in a fixed-tariff context like Sri Lanka's.

Influence on the Proposed Research

Together, these studies confirm that MPC is a well-established and effective method for residential battery dispatch, and it consistently beats simple rule-based control when forecast information is available [1]-[5]. But none of them address a scheduled, known-in-advance outage. [1], [2], [3], and [5] don't consider outages at all, and [4] only addresses unplanned, weather-triggered ones. On top of that, [1], [2], [3], and [5] each assume some kind of infrastructure access, whether shared BESS control, inverter-level MPPT access, or live market telemetry, that most Sri Lankan residential solar households simply don't have. And [4]'s own suggested future work, comparing their MILP-based MPC against a reinforcement-learning-based controller, hasn't been picked up in the literature yet as far as can be determined. These gaps shape the proposed research directly: an MPC-based system built specifically around known scheduled load-shedding, needing no extra hardware, including a user-configurable resilience preference, and adding a secondary DRL comparison to actually test the direction [4] pointed to.

Existing Systems and Products
Commercially, products like the Tesla Powerwall and LG RESU offer automated home battery management, including a storm watch feature that pre-charges the battery ahead of forecast severe weather in supported regions. These systems automate self-consumption optimization and, where supported, charge proactively ahead of forecast extreme weather, using proprietary control software paired with dedicated inverter hardware. They work well for their intended use cases, but availability and functionality depend entirely on the manufacturer's own hardware ecosystem, and they're mainly aimed at homeowners in markets with established grid infrastructure and manufacturer support, like the US or Australia. The limitations are clear for this context: these systems need proprietary hardware that isn't available or supported in Sri Lanka, their anticipatory charging is built around unplanned severe weather rather than scheduled utility load-shedding, and they don't let users adjust the balance between cost savings and resilience, since that behavior is fixed by the manufacturer.

The proposed component addresses these gaps directly. It runs on publicly available schedule and weather data with no proprietary hardware dependency, it's built specifically around scheduled utility load-shedding rather than unplanned weather events, and it introduces a user-configurable resilience preference that neither the academic literature nor the commercial products reviewed here currently offer.

1.3. Research Gap

Existing research on Model Predictive Control (MPC) for residential solar-battery systems addresses cost optimization under normal grid conditions [1], [3], [5], or resilience during unplanned outages caused by extreme weather [4]. None of these studies address battery control for a scheduled, known-in-advance outage, such as the load-shedding enforced by Sri Lanka's Ceylon Electricity Board. This is a meaningfully different problem: the timing of the disruption is already known in advance, rather than needing to be predicted or reacted to after the fact.

In addition, several existing systems assume access to infrastructure not available to most Sri Lankan households shared battery management control [1], inverter-level MPPT access [2], or live electricity market pricing data [3], [5]. This makes these approaches difficult to apply directly to the local context, where residential systems are typically simpler and operate under fixed, government-regulated tariffs rather than dynamic market prices.

The closest related study [4] explicitly identifies the comparison of MILP-based MPC against a reinforcement-learning-based controller as unaddressed future work, since

solving a MILP problem repeatedly can be computationally demanding. To date, this comparison does not appear to have been carried out in the published literature. Furthermore, none of the reviewed systems allow the homeowner to adjust how much the system should prioritize cost savings versus outage resilience behavior is fixed by the underlying optimization design in every case reviewed.

Research Gap Statement:
There is currently no hardware-free battery control system that anticipates scheduled, known-in-advance load-shedding events, allows the homeowner to configure their own balance between cost savings and blackout resilience, and empirically evaluates whether a learning-based (DRL) approach offers any advantage over a calculated optimization (MPC) approach for this specific problem.

2. OBJECTIVE

2.1. Main Objective

Develop and evaluate a hardware free Model Predictive Control system that decides hourly whether a residential solar battery system should self-consume export or reserve power in order to minimize unserved critical load during scheduled outages while remaining cost aware.

2.2. Specific Objectives

- Build a simulation environment that models battery state of charge solar generation household load and the known outage schedule.
- Formulate the battery dispatch problem as a Model Predictive Control optimization with an objective function that balances savings unserved critical load and battery wear.
- Develop and evaluate a rule-based baseline controller to represent existing commercial battery management practice.
- Train a Deep Reinforcement Learning agent as a secondary comparison method to test whether a learning-based approach offers any advantage over the MPC approach.
- Evaluate all three controllers using total hours of unserved critical load and financial savings as the primary performance metrics.

- Implement a user configurable resilience preference that allows the homeowner to adjust the balance between cost savings and blackout resilience.

3. METHODOLOGY

3.1. Research Methodology

The nature of this research question requires testing how a decision system performs across many different weather conditions outage timings and household usage patterns over an extended period. Live deployment on real household inverters is not feasible within the scope of this project since it would require physical hardware access at real customer sites and months of uncontrolled real-world conditions to gather enough outage events for meaningful evaluation. The data available is time series in nature covering solar generation household load battery state and outage schedules which is well suited to a controlled simulation approach rather than a survey or interview-based design. Simulation based research is also the accepted approach in the closely related literature with all five reviewed studies [1] to [5] using simulation to evaluate their proposed MPC controllers before any real-world deployment.

For these reasons this study adopts a simulation based experimental research design. A Model Predictive Control system is developed as the primary decision-making method with a Deep Reinforcement Learning agent implemented as a secondary comparison method. Both methods along with a rule-based baseline are evaluated under identical simulated conditions to allow a fair and repeatable comparison.

3.2. Data, Participants and Experimental Inputs

Data Category	Source	Purpose	Collection Method
PV generation, household load, battery state of charge	Industry partner residential installations	Reflects real household solar and consumption behavior	Provided directly by industry partner
Load shedding schedule	Ceylon Electricity Board (CEB)	Provides the exact outage timing the system anticipates	Scraped from CEB public schedule using a custom script
Weather data (irradiance, temperature, humidity, wind speed)	NASA POWER API	Enables realistic solar generation simulation via pvlib	Retrieved through NASA POWER public API
Battery specifications (capacity, charge and discharge limits)	Industry partner battery datasheets	Defines operating constraints for the MPC formulation	Provided directly by industry partner

Table 1: Data Sources and Experimental Inputs

There are no human participants involved in this study since all experimental inputs are drawn from historical sensor data public schedule data and public weather data. All company provided data is de identified at the site level with no household names or addresses included.

The preparation process involves cleaning and aligning all four data sources to a common hourly time resolution then combining them through a single data loading component so that the simulation environment receives consistent and synchronized inputs.

3.3. Baseline / Benchmark

The proposed Model Predictive Control system is compared against two benchmark approaches.

The first is a rule-based threshold controller which represents current common practice in residential battery management. This controller charges the battery up to a fixed percentage and discharges down to a fixed minimum percentage regardless of upcoming outage timing or forecast conditions. This benchmark is appropriate because it reflects how most existing home battery systems operate today with no anticipatory logic.

The second is a Deep Reinforcement Learning agent trained using either Proximal Policy Optimization or Deep Q Networks. This benchmark is included to empirically test whether a learning-based approach offers any measurable advantage over a calculated optimization approach for this specific problem. This directly addresses future work identified in [4] where the authors note that comparing their MILP based MPC against a reinforcement learning controller had not yet been explored.

Both benchmarks are evaluated under the same simulated data outage schedule and evaluation period as the proposed MPC system to ensure a fair and controlled comparison.

3.4. Evaluation Metrics

Success in this study is measured using metrics that directly reflect the two goals of the proposed system which are blackout resilience and cost awareness.

Metric	Why Relevant	Target / Comparison
Total hours of unserved critical load	Directly measures blackout survival which is the core objective of the system	Lower than rule based baseline and comparable to or better than DRL
Financial savings in LKR	Measures whether resilience is achieved without excessive cost	Comparable to or better than rule based baseline

Performance under forecast error	Tests robustness when PV or load forecasts are inaccurate	Stable performance across error margins of ten and twenty percent

Table 2 : Evaluation Metrics

The first metric is the most important since it directly tests whether the system fulfills its primary purpose of reducing blackout impact. The second metric ensures that improved resilience does not come at an unreasonable financial cost to the homeowner. The third metric tests whether the system remains reliable under realistic forecasting conditions since perfect forecasts are never available in practice.

3.5. Validation Strategy

The validity of the proposed system is demonstrated by running the MPC controller the DRL agent and the rule based baseline over the same simulated year of data under identical outage schedules and weather conditions then comparing their performance on the metrics defined in Table 2. Running all three controllers under identical conditions is the accepted validation practice used in the closely related literature where MPC controllers are consistently benchmarked against baseline and rule based alternatives under the same simulation setup [1] [4] [5].

Additional validation is performed by introducing artificial forecast error into the PV generation and load forecasts at levels of ten and twenty percent. This tests whether the MPC formulation remains robust when forecasts do not exactly match actual conditions which reflects real world deployment since perfect forecasts are never available.

The battery simulation itself is validated by comparing simulated state of charge behavior against the real battery state of charge data provided by the industry partner to confirm that the simulation environment produces realistic and trustworthy results before it is used to evaluate the three controllers.

3.6. Development Approach

An iterative development approach is adopted for this project. This is appropriate because the Model Predictive Control formulation including its objective function constraints and horizon length cannot be fully specified in advance and instead needs to be refined incrementally as simulation results reveal necessary adjustments. Each iteration involves implementing a version of the controller testing it against the baseline and evaluation metrics identifying weaknesses and refining the formulation before moving to the next stage.

The secondary Deep Reinforcement Learning comparison is implemented after the MPC formulation has been sufficiently refined since it depends on the same simulation environment and evaluation setup. This sequencing ensures that the core contribution of the project is stable before time is spent on the secondary comparison component.

This approach is suitable given the exploratory nature of formulating an optimization problem for a real-world energy system where the correct balance between competing objectives such as savings resilience and battery wear is not known in advance and must be discovered through repeated testing and adjustment.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1. Overview of Proposed Solution and its Components

The overall solution is an integrated residential solar intelligence system consisting of four components. The first component analyses rooftop shading and panel geometry to recommend optimal panel placement. The second component assesses solar panel health from photographs to identify degradation. The third component schedules flexible household appliances around forecast solar generation. The fourth component which is the focus of this proposal decides hourly whether the household should self-consume export or reserve solar generated power in order to remain resilient against scheduled outages while staying cost aware.

The proposed component operates using a Model Predictive Control system as its primary decision method with a Deep Reinforcement Learning agent implemented as a secondary comparison method. It takes battery state solar generation forecasts household load forecasts and the known load shedding schedule as input and produces an hourly action recommendation as output.

4.2. System diagram

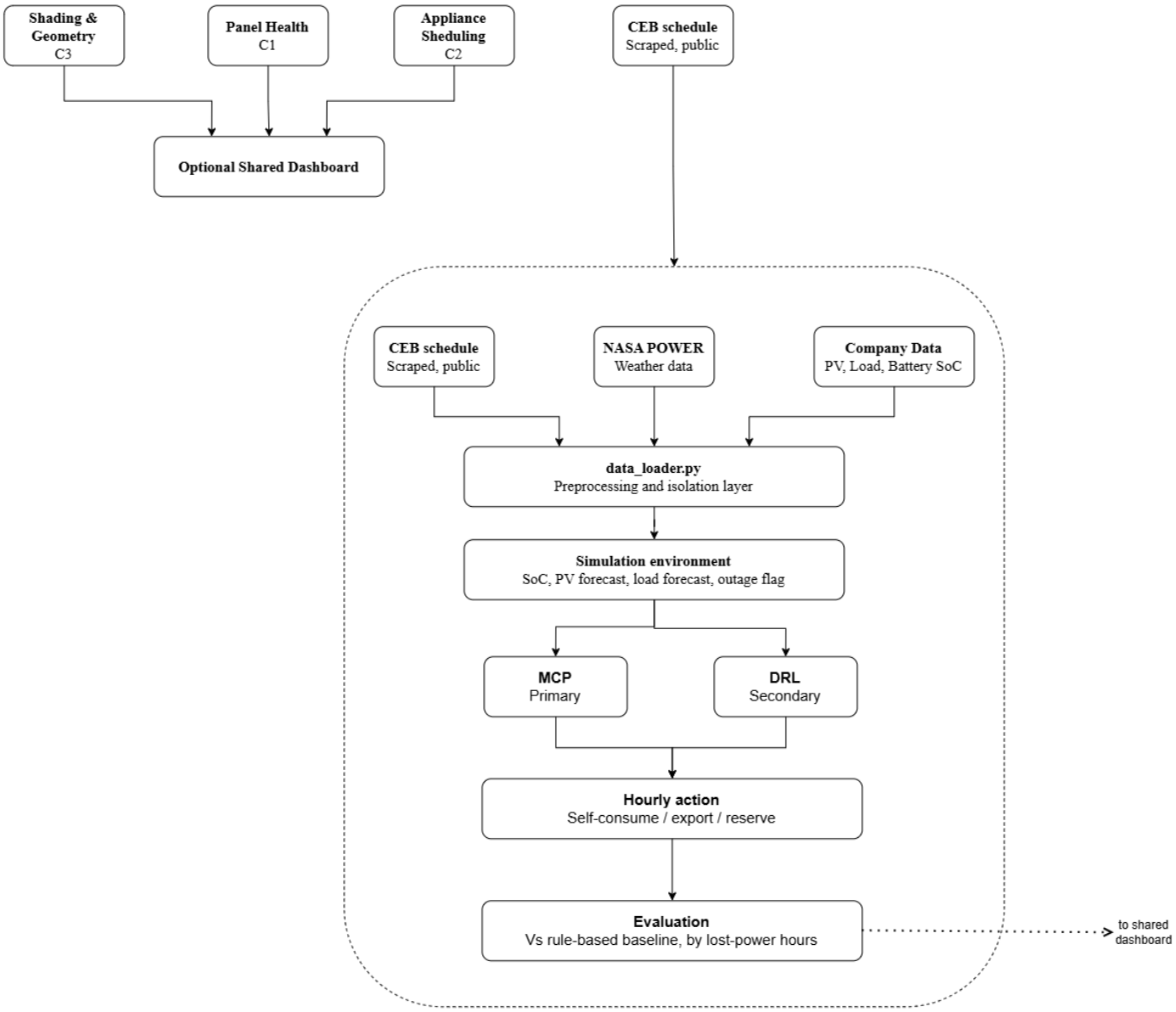


Figure 1: Internal Architecture of Component 4

4.3. Overall Integration

The four components together form a complete residential solar intelligence system, each addressing a different stage of the solar lifecycle from installation through daily operation. The proposed grid survival component operates independently and does not require the output of any other component to function. It receives its own inputs directly from the CEB load shedding schedule NASA weather data and real company generation load and battery data rather than relying on outputs produced by the other three components.

The planned integration method is a shared output format. The hourly action recommendation produced by this component is structured so that it could be displayed alongside the outputs of the other three components in a future shared household dashboard. This integration is optional rather than required since the component is fully functional and independently testable on its own.

This design choice was made deliberately so that the development timeline testing and evaluation of this component are not affected by delays or changes in any other member's work.

4.4. Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
CEB load shedding schedule scraping	✓		
NASA POWER weather data collection	✓		
PV generation simulation using pvlib	✓		
Battery and household load data preparation	✓		
Model Predictive Control formulation	✓		
Deep Reinforcement Learning agent training	✓		

Rule based baseline controller	✓		
Grid survival component evaluation	✓		
Overall project title and problem framing		✓	
Shared dashboard integration format		✓	
Vision based shading and geometry optimization			Senadeera D.M.K.K
Vision based panel health assessment			Erandunu P.H.L
Dynamic solar aware appliance scheduling			Perera P.K.M.P

Table 3: Individual Responsibility Boundary

5. USER REQUIREMENTS

Preliminary requirements evidence was gathered through stakeholder discussion conducted as part of the group's market research into residential solar adopters. This identified that solar system owners currently have no anticipatory response to scheduled outages beyond purchasing additional battery capacity, and that batteries are frequently at a low state of charge exactly when a scheduled outage occurs, since existing systems provide no awareness of the load shedding timetable. A direct structured interview with a solar installer partner is planned as a follow-up activity to validate these findings further during the implementation phase.

5.1. Functional Requirements

No.	Requirement
FR1	The system shall retrieve the current CEB load shedding schedule for the user's specified zone.
FR2	The system shall retrieve current and forecast weather data for the user's location.
FR3	The system shall calculate an hourly recommendation of self consume export or reserve based on battery state forecasts and outage timing.
FR4	The system shall allow the user to configure a resilience preference between prioritizing savings and prioritizing blackout resilience.
FR5	The system shall display the current battery state of charge and the current recommendation to the user.
FR6	The system shall log historical decisions and outcomes for the user to review.

Table 4 : Functional Requirements for Component 4

5.2. Non-Functional Requirements

No.	Category	Requirement
NFR1	Response time	The hourly recommendation shall be computed in under two seconds on standard consumer hardware.
NFR2	Reliability	The system shall continue producing a valid recommendation even if the CEB schedule scraper temporarily fails by falling back to the most recently retrieved schedule.
NFR3	Usability	A new user shall be able to complete initial system setup including tariff and battery configuration in under five minutes.
NFR4	Security	No personally identifying household information shall be stored or transmitted by the system.

NFR5	Scalability	The underlying data pipeline shall support extension to multiple household sites without requiring redesign of the core MPC formulation.
NFR6	Accessibility	The interface shall be usable on both desktop and mobile screen sizes.

Table 5: Non-Functional Requirements for Component 4

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The primary target market is residential rooftop solar owners in Sri Lanka, with residential solar installation and service companies as a secondary market. The primary customer persona is a homeowner who has already invested in a solar-battery system and wants to get more reliable value from that investment, whether through better installation decisions, early detection of panel problems, smarter use of generated power, or protection against scheduled outages.

6.2. Value proposition

The proposed solution is an integrated solar intelligence platform addressing four gaps identified directly from solar owners and installers: siting decisions currently made by rule of thumb with no supporting data, panel degradation that goes unnoticed until it visibly affects the electricity bill, daytime solar generation that goes unused due to inflexible appliance timing, and battery storage that offers no real protection against scheduled outages. The platform brings shading and geometry optimization, panel health assessment, solar-aware appliance scheduling, and outage-resilient energy management together within a single system, rather than requiring separate tools or manual estimation for each concern.

6.3. Revenue Model and Pricing Strategy

A freemium subscription model is proposed for homeowners. The free tier provides basic monitoring and recommendations from each component. A paid tier, estimated at a modest monthly subscription, unlocks advanced features including detailed degradation reports, predictive appliance scheduling, and the full outage-resilience decision engine. A separate business-to-business licensing model is proposed for solar installation companies, offering the platform as a value-added service to their customers through an annual license or API integration fee.

6.4. Cost Estimation

The primary cost areas for this system include software development covering the platform backend and AI model integration, cloud infrastructure for hosting the database and model inference services, external data services for weather and public schedule access, ongoing maintenance covering model updates and customer support, and testing and integration costs associated with validating the system against real installer and household data. These costs are expected to scale gradually with the number of active users and households connected to the platform, with the largest initial investment concentrated in software development and model training, and recurring costs primarily driven by cloud infrastructure and data service usage.

6.5. Competitive Advantage and Intellectual Property

The main competitive advantage is bringing four distinct AI-based capabilities together in a single platform, rather than requiring homeowners or installers to use separate, disconnected tools for installation planning, health monitoring, energy scheduling, and outage resilience. Potential intellectual property includes the trained AI and machine learning models developed for each component, the custom optimization and decision-support algorithms including the grid survival MPC and DRL comparison, the integrated system architecture connecting all four components, and the datasets collected through this research.

7. BUDGET AND JSUTIFICATION

Cost Category	Item	Justification
Hardware costs	None required	The system is designed to be fully hardware-free, requiring no smart meters, inverters, or additional sensors beyond what the household already owns
Software licenses	None required	All core libraries used, including Gymnasium, pvlib, and available MPC solvers, are open source
Cloud services	Compute time for simulation and training	Running repeated MPC simulations across a full year of data and training the secondary DRL agent benefits from faster processing than a personal laptop can reliably provide within the project timeline
Cloud services	Data storage	Storing simulated datasets, model checkpoints, and evaluation results generated across multiple experimental runs
Human effort cost	Researcher time	Time required for data collection and preparation, MPC formulation and tuning, DRL implementation, and evaluation across the project timeline

Table 6: Budget and Justification for C4

The overall cost profile for this component is low compared to typical hardware-based research projects, since the system is intentionally designed to require no additional physical equipment and relies entirely on publicly available data sources alongside data already provided by the industry partner. The main resource requirement is compute time for running simulations and training the DRL comparison agent, and researcher time for the iterative development of the MPC formulation.

8. WORK BREAKDOWN STRUCTURE (WBS)

9. Phase	Task	Responsible	Target Period
Phase 1	Fisheye rooftop image collection	Erandunu P.H.L	Sept – Oct 2026
Phase 1	Tropical degradation image collection	Erandunu P.H.L	Sept – Oct 2026
Phase 1	Day-ahead PV forecasting model development	Perera P.K.M.P	Sept – Oct 2026
Phase 1	CEB schedule scraping, NASA weather data, simulation environment setup	Wijewardena P.K.L (myself)	Sept – Oct 2026
Phase 1	Initial MPC formulation	Wijewardena P.K.L (myself)	Oct 2026, leading into PP1
Phase 1	Obstacle/sky segmentation	Senadeera D.M.K.K	Oct 2026, leading into PP1
Phase 1	Defect/degradation detection model	Erandunu P.H.L	Oct 2026, leading into PP1
Phase 1	MILP appliance scheduling implementation	Perera P.K.M.P	Oct 2026, leading into PP1
Phase 2	Sun-path modelling and tilt/azimuth optimization	Senadeera D.M.K.K	Nov 2026
Phase 2	Panel health scoring and maintenance recommendations	Erandunu P.H.L	Nov 2026
Phase 2	Forecast-deviation monitoring and event-triggered rescheduling	Perera P.K.M.P	Nov – Dec 2026
Phase 2	Rule-based baseline implementation	Wijewardena P.K.L (myself)	Nov 2026

Phase 2	MPC refinement based on early simulation results	Wijewardena P.K.L (myself)	Nov – Dec 2026
Phase 3	Model validation and refinement	Senadeera D.M.K.K	Jan – Feb 2027
Phase 3	Model validation and refinement	Erandunu P.H.L	Jan – Feb 2027
Phase 3	Scheduling model validation and refinement	Perera P.K.M.P	Jan – Feb 2027
Phase 3	DRL secondary comparison training	Wijewardena P.K.L (myself)	Jan – Feb 2027
Phase 3	PP2 preparation, all members	All members	Feb 2027, leading into PP2
Phase 4	Full evaluation and testing	All members, own components	Mar – Apr 2027
Phase 4	Results analysis	All members, own components	Apr – May 2027
Phase 5	Final report writing	All members	May – Jun 2027
Phase 5	Final presentation and viva preparation	All members	Jun 2027
Phase 5	Final submission	All members	By 30 June 2027

Table 7: Work Breakdown Structure

Workload Distribution

All four members are responsible for an equivalent scope of work, each covering data collection, core technical development, and evaluation within their own component. My component's workload is balanced across data engineering (scraping and API integration), optimization formulation (MPC), and a secondary machine learning comparison (DRL), distributing effort across three distinct technical skill areas rather than concentrating entirely on one.

10. GRANTT CHART

Task	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Literature work	X	X								
Research design		X	X							
Data preparation	X	X								
Development		X	X	X						
Experiments					X	X				
Prototype work			X							
Testing							X			
Integration							X			
Evaluation							X	X		
Analysis								X	X	
Report preparation	X	X							X	X

Table 8: Gantt Chart for Component 4

11. REFERENCES LIST

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12. APPENDICES

12.1. Appendix 1: Mandatory Proposal Evidence and Decision Log

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made
Aug 2026	Weather data needed for solar generation simulation must be independently sourced, not shared with teammates	Reviewed available public weather data sources	Requesting shared access to a teammate's weather data pipeline; using only historical company data without external weather input	Independently retrieve weather data through the NASA POWER API to keep the component fully self-contained
Sep 2026	Panel feedback indicated that Deep Reinforcement Learning may not be justified since outage timing is known in advance rather than uncertain	Panel discussion at the proposal presentation; review of five published papers on MPC-based battery management [1]–[5]	Keep DRL as the sole method; remove DRL entirely from the project	Reposition Model Predictive Control as the primary method, with a single DRL agent retained as a secondary comparison
Sep 2026	Needed confirmation that a genuine research gap exists for the revised MPC-based approach	Critical review of five related papers, comparing method, context, and limitations	Assume the gap without checking literature; select a different technology area entirely	Confirmed the gap: no reviewed paper addresses scheduled outages, hardware-free deployment, and an RL-versus-MPC comparison together

Sep 2026	User resilience preference feature needed a clear technical mechanism to be implemented within MPC rather than only DRL	Review of MPC objective function structure across reviewed papers	Only implement the preference feature in the DRL agent; drop the feature entirely	Implement the resilience preference as an adjustable weight within the MPC cost function, kept as the primary method's core feature
Sep 2026	Adopting MPC introduces a methodological overlap with Perera's MILP-based appliance scheduling component, since MILP is a closely related optimization technique	Reviewed Perera's component scope and objective (day-ahead appliance timing) against this component's objective (hourly battery dispatch)	Avoid MPC entirely to prevent overlap; use an unrelated method with no optimization basis	Retain MPC, since the decision variables and objectives are distinct (appliance timing versus battery dispatch); prepared a clear distinction to address if raised during evaluation

Table 9: Proposal Evidence and Decision Log

12.2. Appendix 2: AI Use Disclosure

AI Tool	Purpose of Use	What Was Generated / Assisted	How Output Was Verified	Final Student Contribution
Claude (Anthropic)	Locating and summarizing relevant literature on MPC-based residential battery management	Summarized and critically compared five papers [1]–[5], identifying method, context, findings, and limitations for each	Read and verified each paper's claims directly against the original PDF documents	Selected and sourced all five papers independently prior to this session; wrote the final literature review and gap justification in own words
Claude (Anthropic)	Understanding why Deep Reinforcement Learning was originally selected as the core method	Explanation of DRL's suitability for sequential decision problems with delayed rewards, and discussion of whether it was the right fit for this specific problem	Cross-checked reasoning against the reviewed literature and discussed the assumption directly with supervisor and evaluation panel	Independently evaluated the reasoning and used it to critically question the original method choice rather than accepting it outright

Claude (Anthropic)	Exploring alternative methods after panel feedback questioned the necessity of Deep Reinforcement Learning	Comparison of MPC and DRL for this specific problem, including discussion of feasibility with available data	Confirmed the revised approach through direct discussion with supervisor before finalizing the decision	Made the final decision to adopt MPC as the primary method with DRL retained as a secondary comparison, after supervisor consultation
Claude (Anthropic)	Preparing for the proposal presentation and evaluation panel session	Draft explanations of key technical concepts and a set of anticipated panel questions with sample answers	Practiced answers independently and cross-checked technical explanations against the reviewed literature	Understood and internalized all explanations sufficiently to answer panel questions independently during the actual presentation

Table 10: AI Interaction Log

12.3. Appendix 3: UI sketches

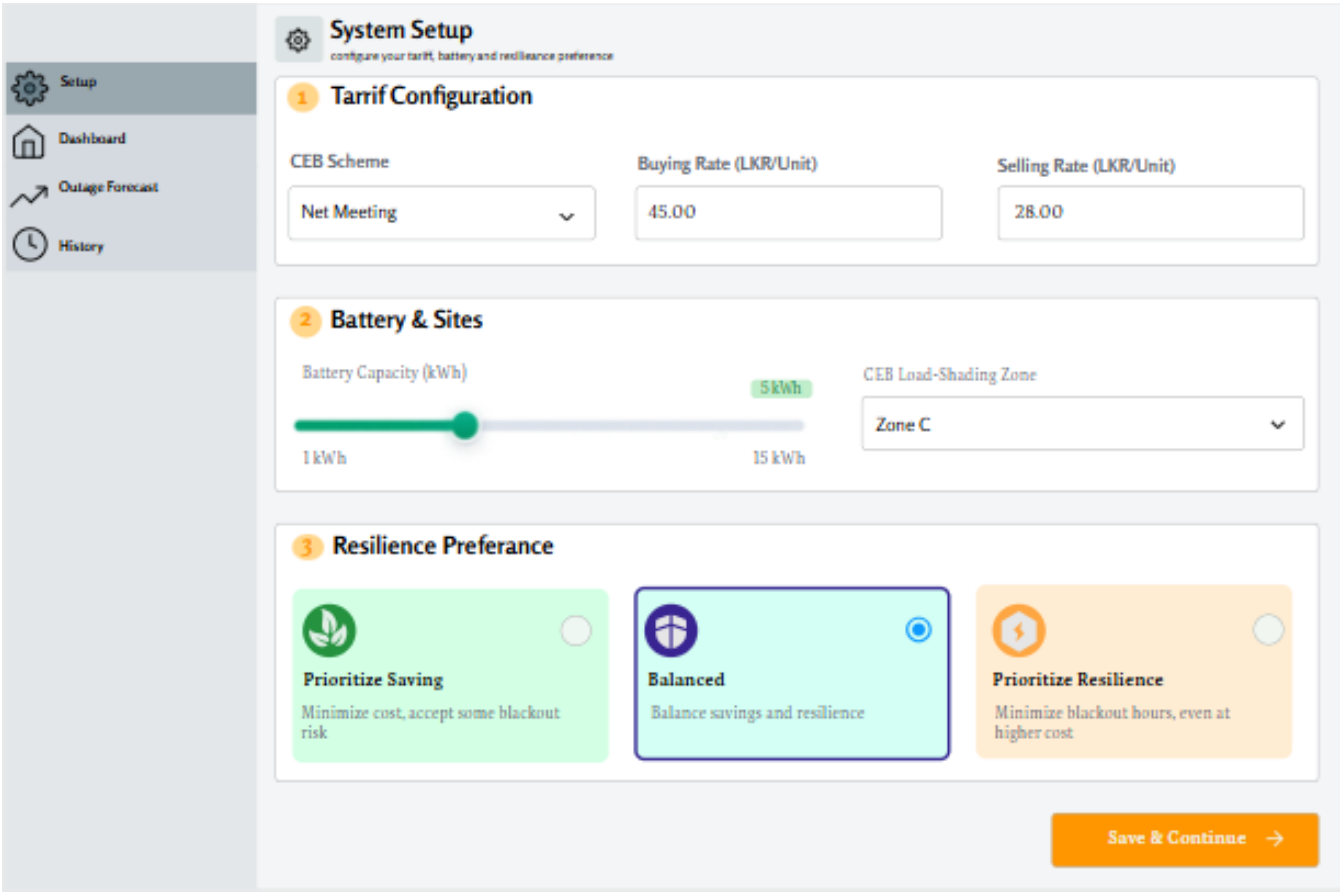


Figure 2: Setup Page

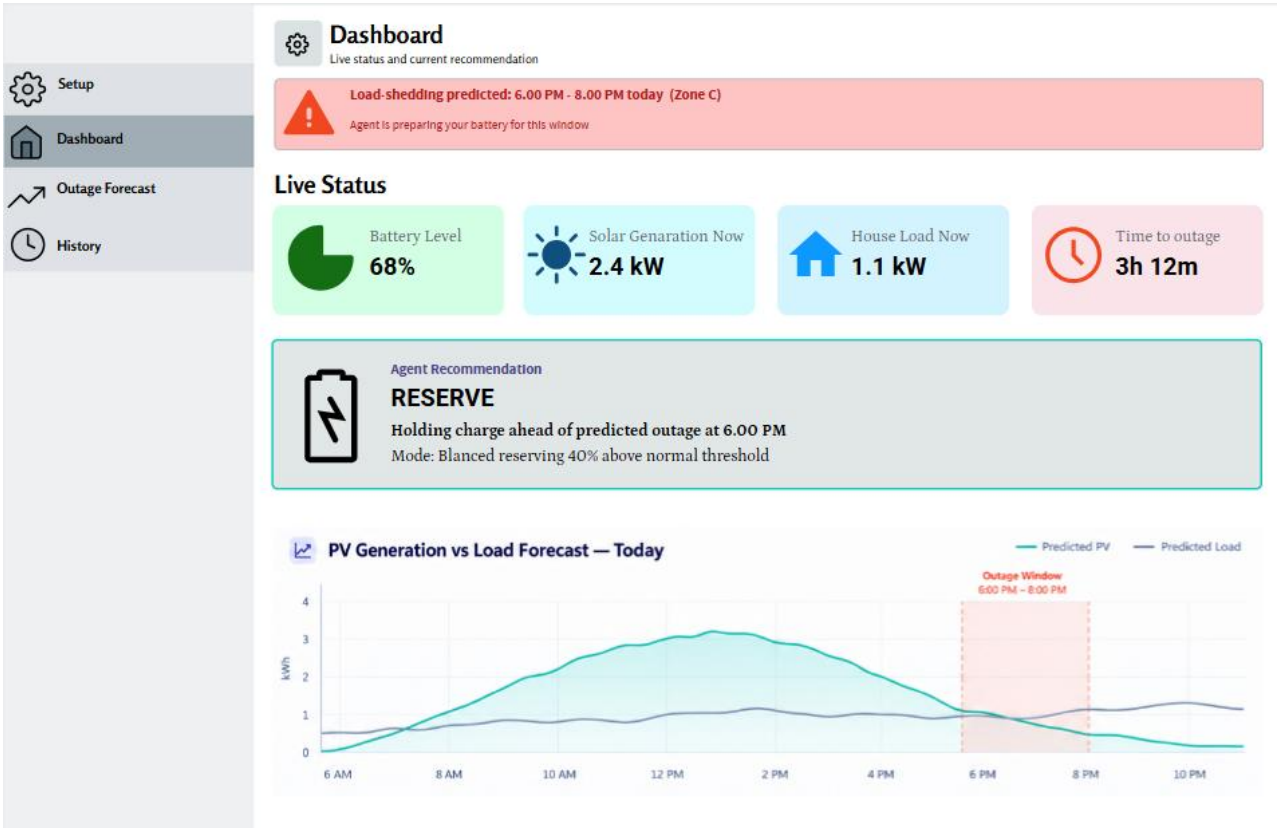


Figure 3: Dash Board

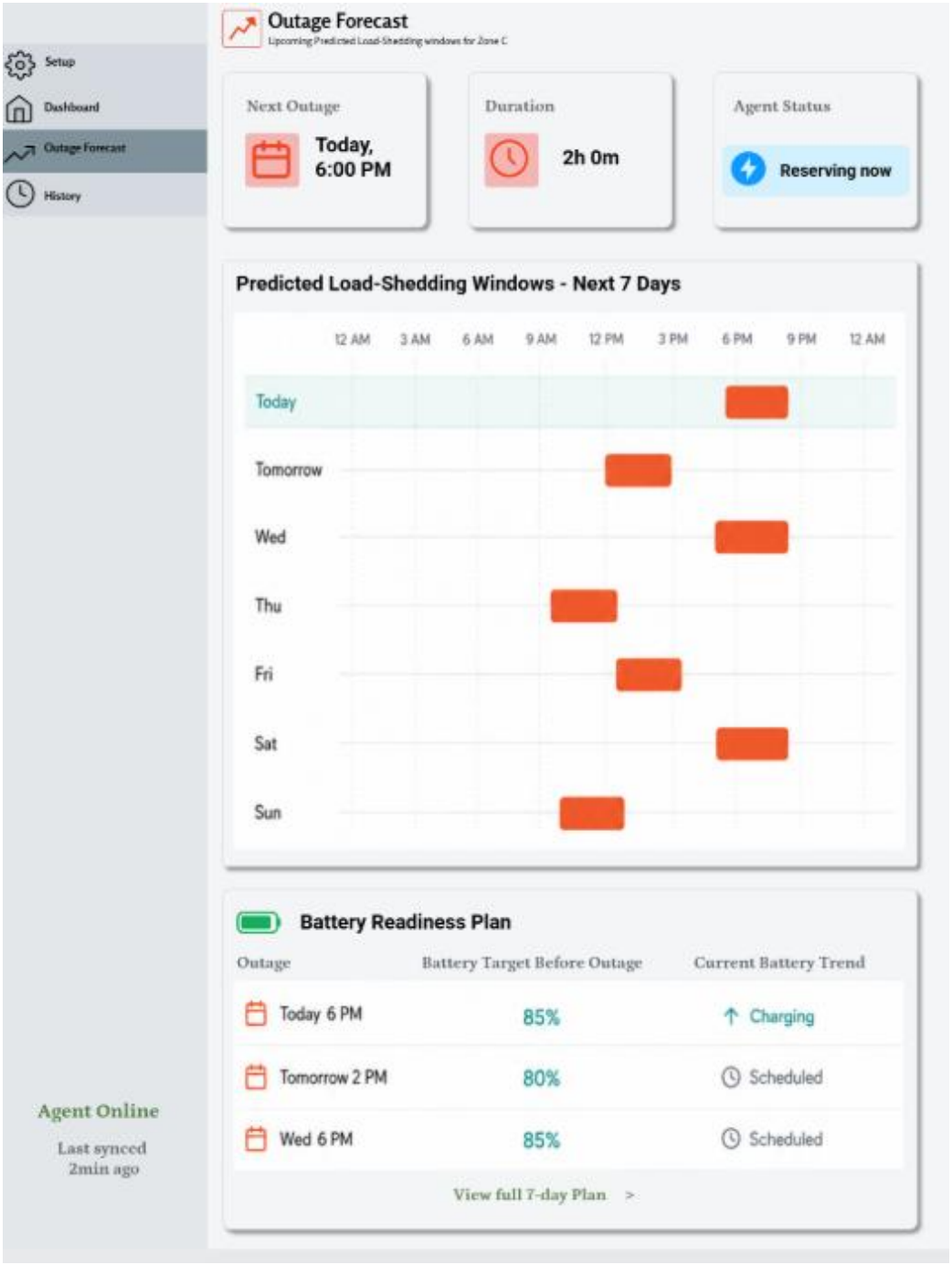


Figure 4: Outage Forecast

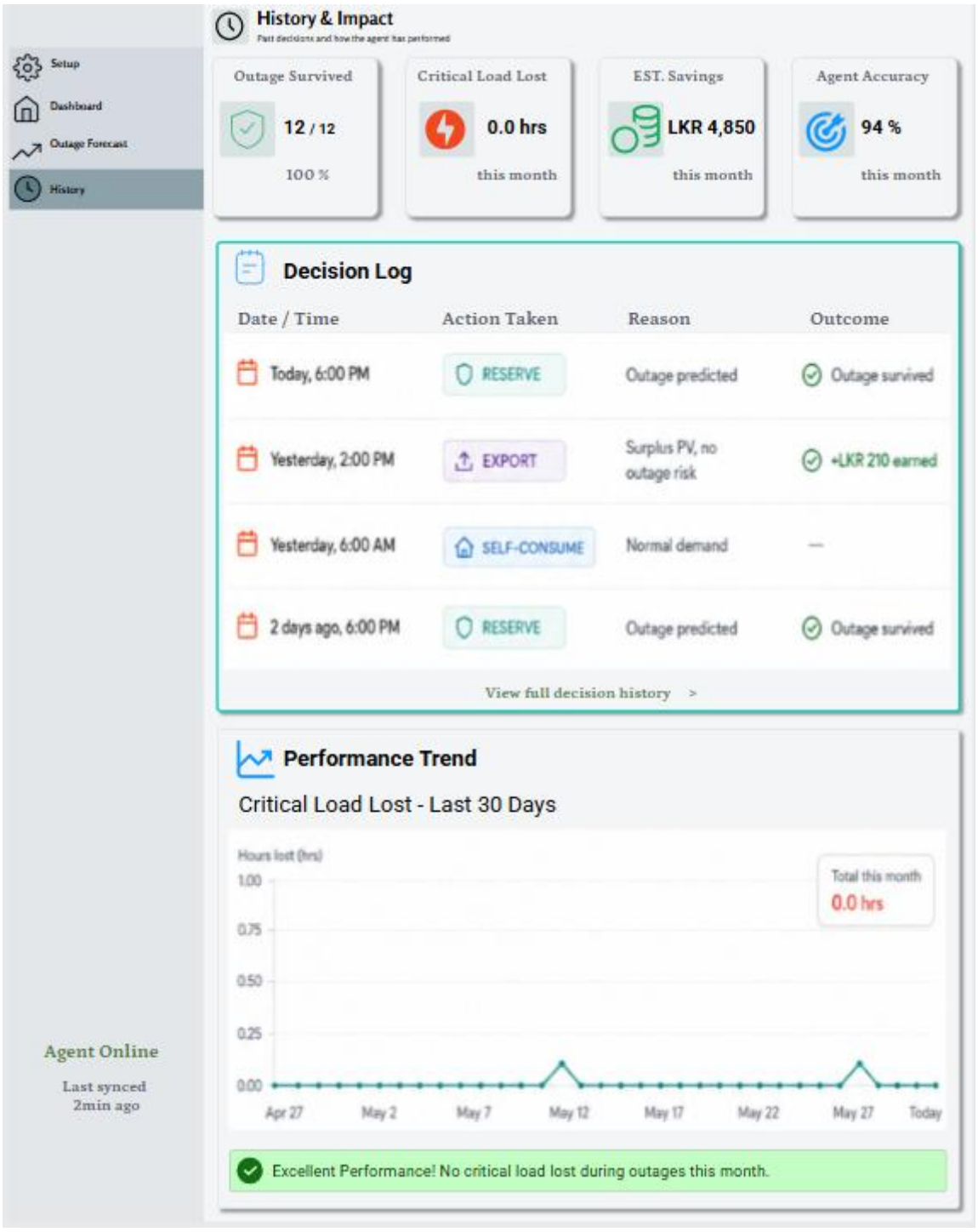


Figure 5: History & Impact